

Phase 2 — Exploratory Data Analysis and Data Quality

Business Goal

Understand hospital operations, financial performance, and data reliability before deploying AI models.

Tasks

- Load data into Python for analysis. Combine them
- Analyze missing values in `approved_amount`, `payment_days`, and `length_of_stay_hours`.
- Perform distribution analysis by department, visit_type, insurance_provider, and city.
- Detect and classify outliers in `billed_amount`, `payment_days`, and `length_of_stay_hours`.
- Engineer features such as visit frequency, average length of stay per patient, provider rejection rate, days since registration, and time-based features.

Deliverables

- EDA notebook (01_eda.ipynb)
- Feature engineering script (build_features.py)
- Modeling dataset (model_table.csv or parquet)
- Data quality report (docx or markdown)